Topic 2: Introduction to Data Understanding

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1 Introduction to Data Understanding

Data understanding is the process of exploring and analyzing collected data to grasp its characteristics, structure, and quality. This step is essential before diving into deeper analysis, predictions, and decision-making.

Importance of Data Understanding

Understanding data is crucial for several reasons:

1. **Informed Decision-Making**: Proper understanding leads to informed business decisions, which can significantly impact an organization's strategy and operations.
2. **Identifying Patterns**: It helps in identifying trends, patterns, and anomalies within the dataset, which can be critical for predictive analytics.
3. **Validity of Insights**: A comprehensive understanding ensures that insights derived from data are valid and applicable, mitigating biases and inaccuracies.
4. **Data Quality Assessment**: Through data understanding, one can evaluate data quality, revealing potential issues in accuracy, relevance, and completeness.

Example

- A marketing team analyzing customer data can identify purchasing patterns that lead to enhanced targeted advertising strategies.
- A healthcare provider using patient data can spot trends in disease outbreaks, enabling proactive measures.

2 Measures of Central Tendency

Measures of central tendency are critical statistics in understanding the distribution and characteristics of a dataset. These measures provide a means of summarizing and describing the center of a dataset, encapsulating a representative value that embodies the data as a whole. The three most common measures of central tendency are the mean, median, and mode.

2.1 Mean: Definition and Calculation

Definition: The mean, often referred to as the arithmetic average, is calculated by adding all values in a dataset and dividing by the total number of values.

Calculation:

Let $x_1, x_2, \ldots, x_n$ be the values in a dataset.

The formula for the mean μ is given by:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Example: Consider the dataset $\{5, 10, 15, 20, 25\}$.

Calculating the mean:

$$\mu = \frac{1}{5}(5 + 10 + 15 + 20 + 25) = \frac{75}{5} = 15$$

2.2 Median: Definition and Calculation

Definition: The median represents the middle value when a dataset is ordered from least to greatest. If the dataset is even in size, the median will be the average of the two middle numbers.

Calculation:

1. Arrange the data in ascending order.
2. If n is odd, the median is:

$$\text{Median} = x_{\left(\frac{n+1}{2}\right)}$$

3. If n is even, use:

$$\text{Median} = \frac{x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}}{2}$$

Example: In the dataset $\{12, 15, 10, 13\}$ (sorted $\{10, 12, 13, 15\}$):

$$\text{Median} = \frac{12 + 13}{2} = 12.5$$

2.3 Mode: Definition and Calculation

Definition: The mode is the value that appears most frequently in a dataset. A dataset can be unimodal (one mode), bimodal (two modes), or multimodal (more than two).

Calculation:

Identify the frequency of each value, and the one with the maximum frequency is the mode.

Example : In $\{1, 2, 2, 3, 4\}$:

$$\text{Mode} = 2$$

2.4 When to Use Each Measure and Real-World Examples

- **Mean:** The mean is useful for datasets with no outliers since it can be heavily affected by extreme values. For example, calculating the average salary in a company may give a good overall picture. However, if a few executives earn much more than others, it may misrepresent the majority's salary.

- **Median:** The median is preferable in skewed distributions or when outliers are present. For instance, in determining the median home price in a neighborhood where most homes are affordable, but a few luxury homes inflate the mean, the median provides a more accurate reflection of what most buyers might expect.
- **Mode:** The mode is useful in categorical data to understand the most common category. For example, in market research, if a survey identifies preferences for types of products, knowing which product has the highest frequency (mode) can influence marketing strategy.

3 Mesures of Variability

Variability is a crucial concept in statistics that reflects the spread or dispersion of a set of data points. It helps us understand how much the data points differ from each other and from the average value, thus providing valuable insights.

3.1 Range: Definition and Calculation

The **range** is the simplest measure of variability. It is defined as the difference between the maximum and minimum values in a dataset.

Formula

$$\text{Range} = \text{Max}(X) - \text{Min}(X)$$

Calculation:

1. Identify the maximum and minimum values in the data set.
2. Subtract the minimum from the maximum.

Example

Consider the dataset $X = \{4, 8, 15, 16, 23, 42\}$.

1. Max value = 42
2. Min value = 4
3. Range $= 42 - 4 = 38$

3.2 Variance: Definition and Calculation

Variance quantifies the degree of spread in a dataset. It is the average of the squared differences from the mean.

Formula

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (X_i - \mu)^2$$

where:

- σ^2 = variance
- μ = mean of the dataset
- N = number of observations

Calculation Steps

1. Calculate the mean μ of the dataset.
2. Subtract the mean from each data point and square the result.
3. Sum these squared differences.
4. Divide by the total number of observations N.

Example

Consider the dataset $X = \{5, 6, 8\}$.

1. Mean $\mu = \frac{5+6+8}{3} = 6.33$
2. Deviations: $5 - 6.33 = -1.33$, $6 - 6.33 = -0.33$, $8 - 6.33 = 1.67$
3. Squared deviations: $(-1.33)^2 = 1.7689$, $(-0.33)^2 = 0.1089$, $(1.67)^2 = 2.7889$
4. Sum of squared deviations $= 1.7689 + 0.1089 + 2.7889 = 4.6667$
5. Variance $= \frac{4.6667}{3} \approx 1.5556$

3.3 Standard Deviation: Definition and Calculation

The **standard deviation** is the square root of the variance and provides a measure of variability in the same units as the data, making it more interpretable.

Formula

$$\sigma = \sqrt{\sigma^2}$$

Calculation Steps

1. Calculate the variance as shown above.
2. Take the square root of the variance to find the standard deviation.

Example

Using the variance calculated above ($\sigma^2 \approx 1.5556$), the standard deviation is:

$$\sigma \approx \sqrt{1.5556} \approx 1.25$$

3.4 Importance of Understanding Variability with Real-Life Applications

Understanding variability is essential in many fields. It helps make informed decisions based on data analysis, assess risks, and comprehend patterns in various phenomena.

1. **Finance:** In investing, variability indicates risk. For example, a stock with high volatility (standard deviation) is considered riskier, impacting investment decisions.
2. **Quality Control:** In manufacturing, a product's quality is often assessed through its variability. A high variance in product measurements might indicate inconsistent production processes.
3. **Healthcare:** In medical studies, understanding the variability in patient responses to treatment enables better tailored therapies for individual needs.
4. **Social Sciences:** Variability in survey results can reveal insights about diverse population responses and behavior patterns.
5. **Sports Analytics:** Analyses of athletes' performance with respect to variability help coaches devise better training regimes based on performance consistency.

4 Distribution Shapes

Understanding the different shapes of distributions is essential for data analysis, statistical modeling, and the interpretation of results. We will discuss normal distribution, skewness, kurtosis, and their practical implications.

4.1 Normal Distribution

The normal distribution, often referred to as Gaussian distribution, is a fundamental concept in statistics. It is characterized by its bell-shaped curve, symmetric around the mean.

Characteristics

1. **Mean, Median, and Mode**: In a normal distribution, these three measures of central tendency are equal.
2. **Symmetry**: The left and right sides of the curve are mirror images.
3. **Standard Deviation**: The spread of data points from the mean is defined by the standard deviation (σ). About 68% of data points lie within one standard deviation, 95% within two, and 99.7% within three (Empirical Rule).
4. **Mathematical Definition**: The probability density function (PDF) of a normal distribution is given by the equation:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = mean
- σ = standard deviation
- e = 2.72

Numerical Examples:

1. **Example 1**: A dataset has a mean (μ) of 50 and a standard deviation (σ) of 10. What proportion of data falls between 40 and 60?
   - This range corresponds to (μ - σ) and (μ + σ). According to the Empirical Rule, 68% of data falls within this range.
2. **Example 2**: For height data of adults (mean = 170 cm, standard deviation = 5 cm), what percentage of individuals are taller than 180 cm?
   - Calculate the z-score: $z = (180 - 170)/5 = 2$.
   - Using z-tables, the area to the left is approximately 0.9772; thus, (1 - 0.9772 = 0.0228) or 2.28% are taller than 180 cm.

Importance:

The normal distribution supports many statistical methods, including hypothesis testing, confidence intervals, and regression analysis due to the Central Limit Theorem. The Central Limit Theorem states that the sample average of a large number of variables independently drawn from the same distribution is approximately normally distributed, regardless of the original distribution of the variables.

4.2 Skewness

Skewness measures the asymmetry of a distribution.

Characteristics:

1. **Positive Skew**: The tail on the right side is longer or fatter. The mean is greater than the median.
2. **Negative Skew**: The tail on the left side is longer or fatter. The mean is less than the median.

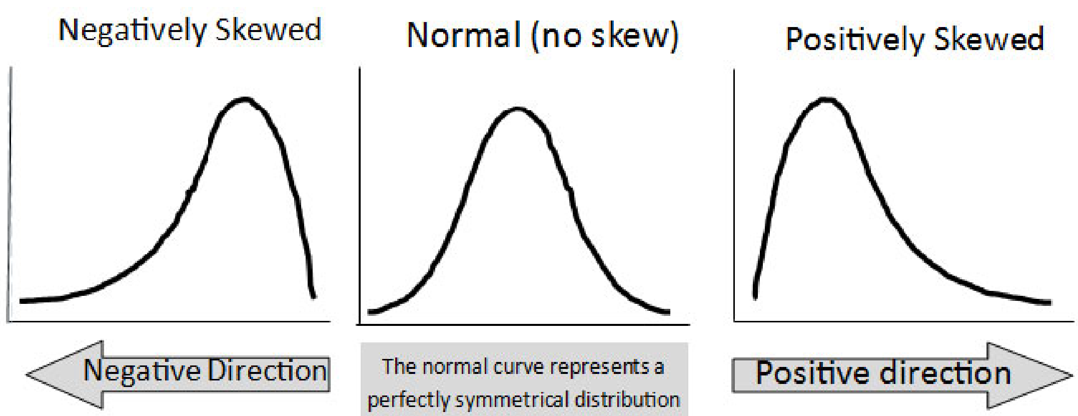

Figure 1: Measure of Skewness

4.3 Kurtosis

Kurtosis measures the "tailedness" of a distribution. It informs how outlier-prone a distribution is.

Types of Kurtosis

1. **Mesokurtic**: Normal distribution with kurtosis = 3.
2. **Leptokurtic**: More peaked than normal (kurtosis > 3), indicating heavy tails and more outliers.
3. **Platykurtic**: Flatter than normal (kurtosis < 3), indicating light tails.

4.4 Real-World Examples

1. **Normal Distribution**: Heights of adults, errors in measurements, and standardized test scores often follow a normal distribution.
2. **Positive Skew**: Income distribution is typically right-skewed, with most people earning below the mean due to a few high earners.
3. **Negative Skew**: Age at retirement might show a negative skew as many people retire around the same age, but a few may retire much later.

5 Data Visualization Using Plots

5.1 Introduction

Data visualization is the graphical representation of information and data. By using visual elements such as charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

Key reasons for utilizing data visualization include:

- **Simplification**: Complex data sets can be simplified to convey the right message clearly.
- **Pattern Recognition**: Facilitates identifying trends and correlations that might not be apparent in raw data.
- **Decision Making**: Visualized data supports informed decision-making by illustrating significant insights clearly.

Effective data visualization requires an understanding of the data being represented, choosing the appropriate plot type, and effectively communicating the message to the audience.

5.2 Bar Charts

Bar charts are one of the simplest forms of data visualization, especially useful for comparing quantities across different categories.

Purpose

- Compare discrete categories of data.
- Easily depict the magnitude of values associated with each category.
- Highlight differences between categories quickly.

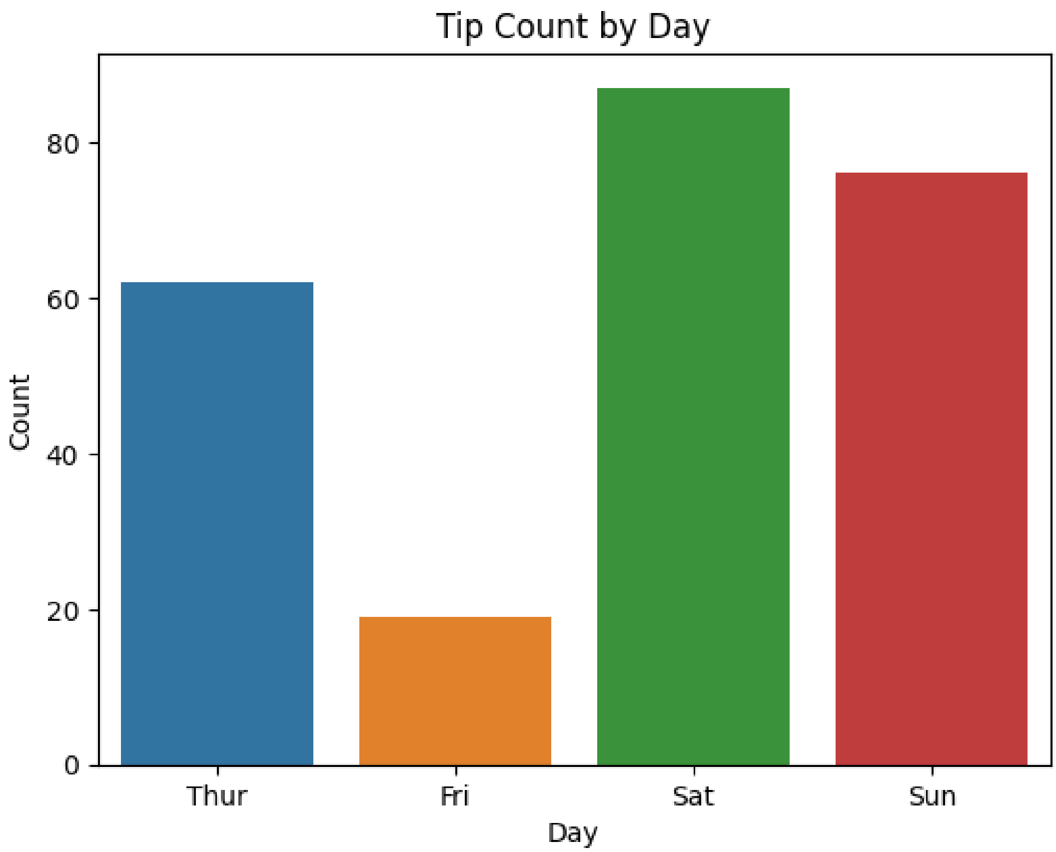

Figure 2: alt text

This chart visually shows that Friday has the least number of tips.

5.3 Histograms

Histograms are used to visualize the distribution of numerical data by dividing it into bins or intervals.

Purpose

- Understand the underlying frequency distribution of the data.
- Identify the central tendency, variability, and skewness of data.

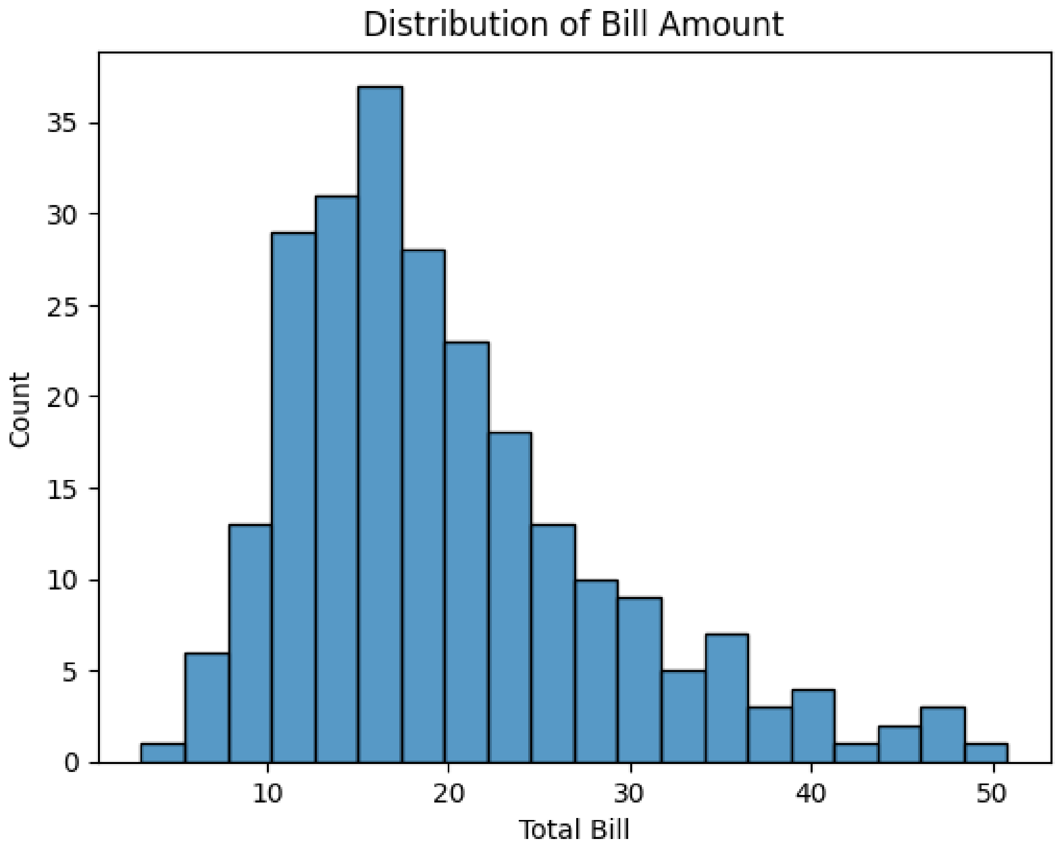

Figure 3: alt text

5.4 Box Plots

Box plots, or whisker plots, visually summarize the distribution of a dataset based on a five-number summary: minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum.

- **Q1 (First Quartile)**: 25th percentile, value below which 25% of the data falls.
- **Q2 (Median)**: 50th percentile or median value.
- **Q3 (Third Quartile)**: 75th percentile, value below which 75% of the data falls.

Outliers are typically defined as observations that fall below (Q1 - 1.5 $\times$ IQR) or above (Q3 + 1.5 $\times$ IQR), where IQR (Interquartile Range) = (Q3 - Q1).

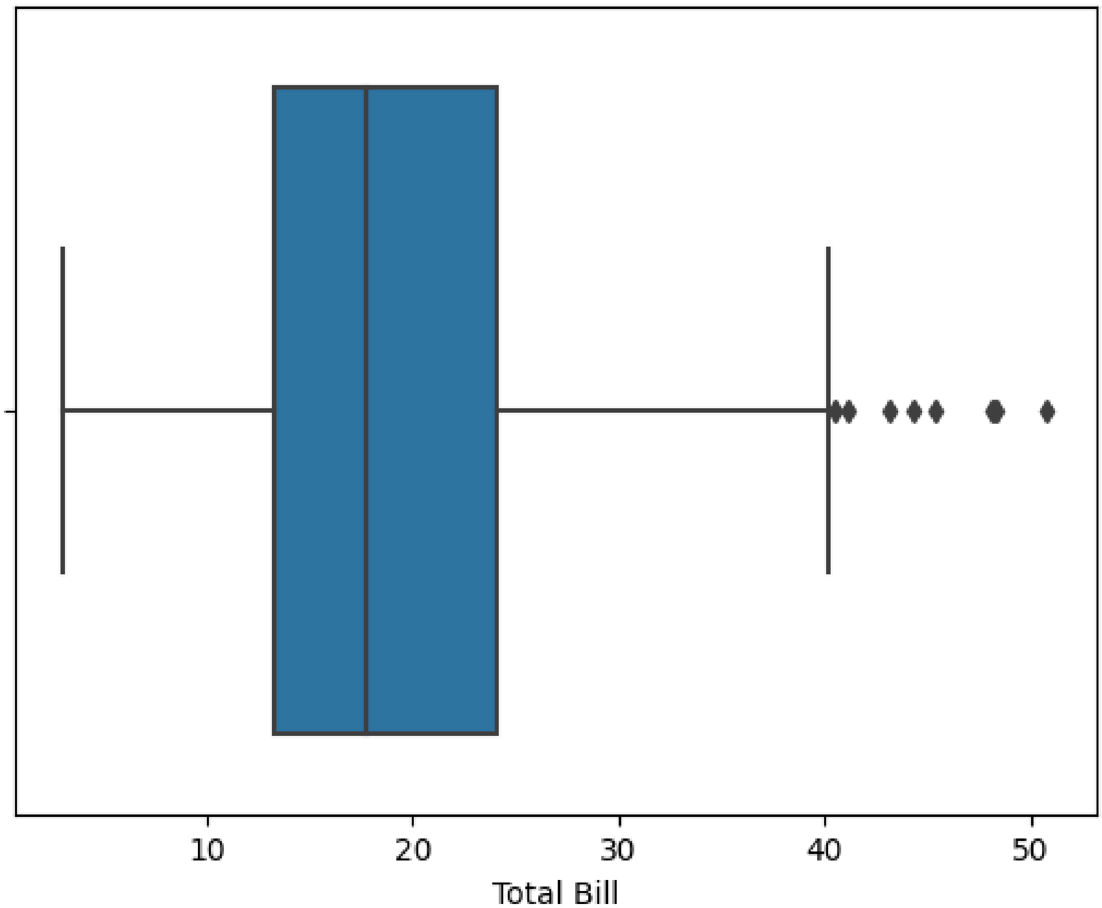

Figure 4: Box Plot

5.5 Scatter Plots

Scatter plots are invaluable for visualizing relationships between two quantitative variables by representing data points in two dimensions. It helps to:

- Identify the type of relationship (positive, negative, or no correlation).
- Highlight trends and potential outliers.

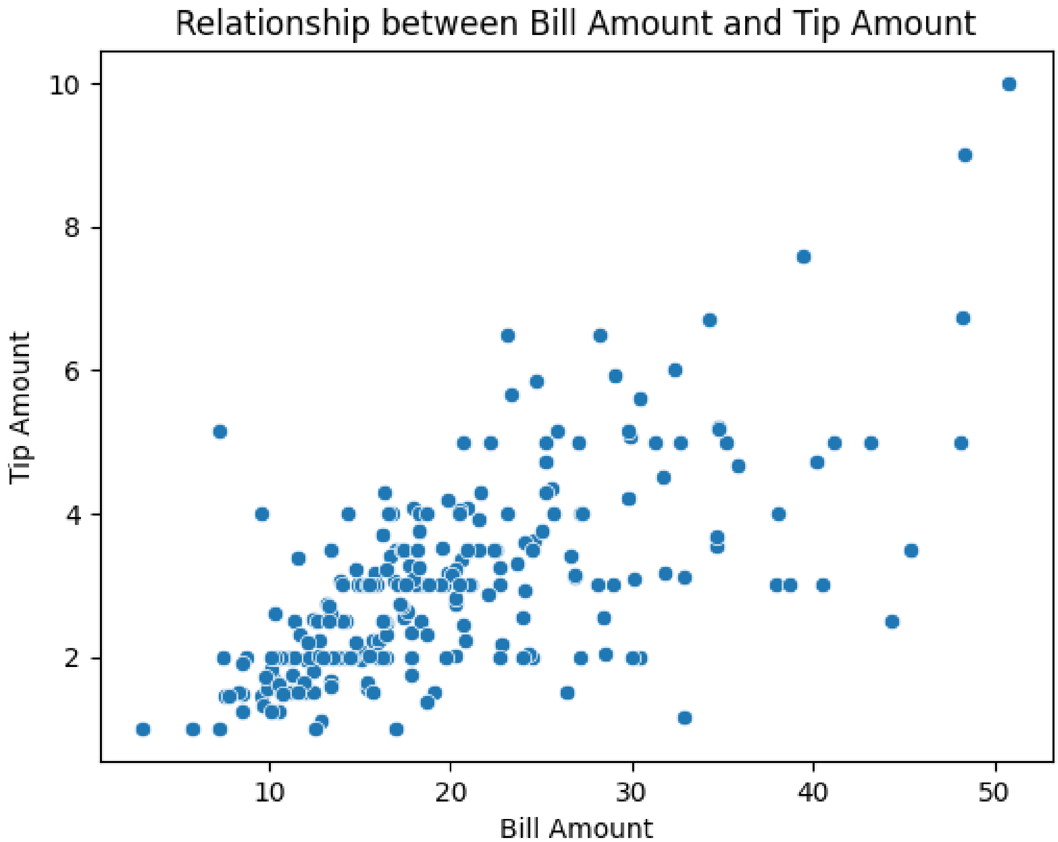

Figure 5: Scatter Plot

6 Correlation

6.1 Definition

Correlation quantifies the degree to which two variables are related. Mathematically, correlation is a coefficient that ranges from -1 to +1:

- A correlation of +1 indicates a perfect positive linear relationship between the variables.
- A correlation of -1 indicates a perfect negative linear relationship.
- A correlation of 0 suggests no linear relationship between the variables.

The formula for correlation can be represented as follows:

$$r = \frac{N(\sum xy) - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2][N\sum y^2 - (\sum y)^2]}}$$

Where:

- r is the correlation coefficient,
- N is the number of data points,
- x and y are the individual sample points.

6.2 Positive vs. Negative Correlation

- **Positive Correlation**: This occurs when an increase in one variable corresponds with an increase in another variable. For instance, height and weight often show a positive correlation; as height increases, weight typically increases too.
- **Negative Correlation**: This occurs when an increase in one variable corresponds with a decrease in another. An example more exercise often leads to lower body weight.

Remember, correlation does not imply causation; hence, caution should be exercised in drawing conclusions about the nature of relationships based solely on correlation coefficients.

7 Covariance

7.1 Definition

Covariance is a statistical measure that indicates the extent to which two random variables change together. It reflects the degree to which one variable tends to increase (or decrease) when the other variable does. Covariance can be mathematically defined as follows:

$$Cov(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

where ($\bar{x}$) is the sample mean of (X) and ($\bar{y}$) is the sample mean of (Y).

Example: Suppose we have the following data points for random variables (X) and (Y):

$$X = \{2, 4, 6\}, \quad Y = \{1, 3, 5\}$$

First, calculate the means:

$$\bar{x} = \frac{2+4+6}{3} = 4, \quad \bar{y} = \frac{1+3+5}{3} = 3$$

Now, calculate the covariance:

$$Cov(X, Y) = \frac{1}{3-1} [(2-4)(1-3) + (4-4)(3-3) + (6-4)(5-3)]$$

$$= \frac{1}{2} [(-2)(-2) + 0 + (2)(2)] = \frac{1}{2} [4 + 0 + 4] = \frac{8}{2} = 4$$

7.2 Interpreting Covariance

The sign of the covariance is critical in interpreting the relationship between the two variables:

1. **Positive Covariance**: Indicates that as one variable increases, the other variable tends to increase as well. This suggests a direct relationship.
2. **Negative Covariance**: Indicates that as one variable increases, the other variable tends to decrease. This suggests an inverse relationship.
3. **Zero Covariance**: Suggests that there is no linear relationship between the variables.

8 Cross Tabulation

Cross tabulation, also known as contingency table analysis, is a powerful statistical tool used to analyze the relationship between two or more categorical variables. It allows to observe and interpret the frequency distributions of variables. The main objectives of cross tabulation are:

1. To summarize large data sets in a manageable form.
2. To identify potential relationships or patterns between variables.
3. To assist in hypothesis testing and inferential statistics.

Cross tabulation involves the creation of a matrix (or table) that displays the frequency distribution of variables. Each cell within the table represents the intersection of a row and a column variable, showing how often each combination occurs.

The contingency table can be represented as:

$$\begin{array}{c|c|c|c|c} & Y_1 & Y_2 & \cdots & Y_m \\ \hline X_1 & n_{11} & n_{12} & \cdots & n_{1m} \\ X_2 & n_{21} & n_{22} & \cdots & n_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ X_k & n_{k1} & n_{k2} & \cdots & n_{km} \end{array}$$

Where

- Two categorical variables: X (with k levels) and Y (with m levels).
- Let n_{ij} be the frequency count for the i-th level of X and the j-th level of Y.

Hence, treating outliers may involve reassessing the data collection methods, employing transformations, or excluding outliers with justification to enhance the integrity of the analysis. Understanding, identifying, and managing outliers is a key skill in statistical analysis, imperative for obtaining reliable results and insights.

9 Missing Values

Missing values occur when data is absent in a dataset, meaning certain observations have no recorded value for one or more features.

Example

Consider a dataset of student grades.

Student	Math	Science	English
A	90	85	NaN
B	NaN	75	88
C	80	NaN	93
D	85	90	87

The missing values can be observed clearly in the table. In this case, we notice one NaN value in English for Student A, one in Math for Student B, and one in Science for Student C.

Use functions such as `isnull().sum()` in Pandas to get a quick overview of missing values.

```python
import pandas as pd

data = {'Student': ['A', 'B', 'C', 'D'],
        'Math': [90, None, 80, 85],
        'Science': [85, 75, None, 90],
        'English': [None, 88, 93, 87]}

df = pd.DataFrame(data)
print(df.isnull().sum())
```

9.1 Causes of Missing Data

- **Human Error** – Mistakes in data entry or survey responses.
- **Data Collection Issues** – Sensor failures, lost records, or skipped survey questions.
- **Privacy Restrictions** – Some data might be intentionally left blank.
- **Irrelevant Data** – Certain values don't apply to all cases (e.g., salary for unemployed people).

9.2 Types of Missing Data

1. **Missing Completely at Random (MCAR)** – No pattern in missingness; purely random.

2. **Missing at Random (MAR)** – Missing data depends on other observed data but not the missing value itself.

3. **Missing Not at Random (MNAR)** – Missing values depend on unobserved data, often introducing bias.

Missing Data Type	Why Data Is Missing	Example
MCAR	Pure randomness; unrelated to data	Survey glitch deletes some GPA responses
MAR	Missingness depends on **observed** variables	Business majors skip GPA more often
MNAR	Missingness depends on **unobserved** value itself	Low-GPA students avoid reporting GPA

10 Outlier Detection

Identifying values that differ significantly from the rest of the dataset is crucial, as they can impact statistical analyses. Here are some consequences:

- **Mean**: Outliers can disproportionately affect the mean, inflating or deflating it.
- **Standard Deviation**: The standard deviation is sensitive to outliers, leading to a possible misrepresentation of variability.
- **Correlation**: In regression analysis, outliers can strongly influence the slope of the regression line, potentially leading to misleading conclusions.

10.1 Z-score Method

$$z = \frac{x - \mu}{\sigma}$$

Where (x) is the observation, (μ) is the mean, and (σ) is the standard deviation.

A common threshold for determining outliers using Z-scores is $|Z| > 3$.

Example:

Consider a dataset: ([10, 12, 12, 13, 12, 13, 20]).

1. Calculate the mean (μ):

$$\mu = \frac{10 + 12 + 12 + 13 + 12 + 13 + 20}{7} = 12.857$$

2. Calculate the standard deviation (σ):

$$\sigma = \sqrt{\frac{\sum(X_i - \mu)^2}{N}} = \sqrt{\frac{(10 - 12.857)^2 + (12 - 12.857)^2 + ... + (20 - 12.857)^2}{7}} \approx 2.9$$

3. Calculate Z-score for (X = 20):

$$Z_{20} = \frac{(20 - 12.857)}{2.96} \approx 2.44$$

Since $|Z_{20}| < 3$, 20 is not an outlier by this method.

10.2 IQR Method

The IQR method uses the interquartile range to identify outliers. IQR is defined as:

$$IQR = Q3 - Q1$$

Where (Q1) is the first quartile (25th percentile) and (Q3) is the third quartile (75th percentile). Outliers are identified by the criterion:

$$\text{Outlier if } X < (Q1 - 1.5 \times IQR) \text{ or } X > (Q3 + 1.5 \times IQR)$$

Example:

Using the same data set ([10, 12, 12, 13, 12, 13, 20]):

1. Find (Q1) (first quartile) and (Q3) (third quartile):
   - (Q1 = 12)
   - (Q3 = 13)

2. Calculate (IQR):

$$IQR = Q3 - Q1 = 13 - 12 = 1$$

3. Determine the lower and upper bounds:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR = 12 - 1.5 \times 1 = 10.5$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR = 13 + 1.5 \times 1 = 13.5$$

4. Identify outliers:

   - (20) exceeds (13.5) and thus is considered an outlier.

Numerical Example:

Consider two datasets:

1. ([10, 12, 12, 14]) yields a mean of (12.0) and a standard deviation of (1.0).
2. ([10, 12, 12, 100]) has a mean of (33.5) and standard deviation (43.63).

The impact of the outlier ((100)) dramatically alters both measures, demonstrating the importance of outlier detection.

11 Different Statistical Tests

11.1 T-Tests: Independent and Paired Samples

T-tests assess whether the means of two groups are statistically different from each other. There are two main types: independent samples t-test and paired samples t-test.

- **Null Hypothesis (H_0)**: Two groups have equal means (no significant difference).
- **Alternative Hypothesis (H_1)**: Two groups does not have equal means.

Formula:

The t-statistic is calculated as follows:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where:

- ($\bar{X}_1$, $\bar{X}_2$) are sample means
- (s_p) is the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- (n_1, n_2) are the sample sizes
- (s_1, s_2) are the sample standard deviations

Example:

Consider two teaching methods (Method A and Method B) tested on student performance. Group A has (n_1=30) students with ($\bar{X}_1$=85), (s_1=10) and Group B has (n_2=30) with ($\bar{X}_2$=78), (s_2=12):

1. Calculate (s_p):

$$s_p = \sqrt{\frac{(30-1)(10^2) + (30-1)(12^2)}{30 + 30 - 2}} = \sqrt{\frac{(29)(100) + (29)(144)}{58}} \approx 11.04$$

2. Calculate (t):

$$t = \frac{85 - 78}{11.04\sqrt{\frac{1}{30} + \frac{1}{30}}} \approx 2.45$$

```python
from scipy.stats import t

# Given values
t_value = 2.45
df = 58  # degrees of freedom (n_1 + n_2 - 2)

# Calculate the two-tailed p-value
p_value = 2 * (1 - t.cdf(t_value, df))

print(f"P-value: {p_value:.4f}") #0.0173
```

Since p-value is less than 0.05, two groups does not have equal means. ## Paired Samples T-Test

The paired t-test is a statistical method used to compare the means of two related groups. It is commonly applied when the same subjects are measured twice, such as before and after a treatment:

- **Null Hypothesis (H_0)**: The mean difference between the paired observations is zero.
- **Alternative Hypothesis (H_1)**: The mean difference between the paired observations is not zero.

Formula:

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}$$

Where:

- ($\bar{D}$) is the mean of the differences
- (s_D) is the standard deviation of the differences
- (n) is the number of paired observations

Example:

If you measure the weight of individuals before and after a diet plan for (n=10) individuals with mean difference ($\bar{D}$=2.5) and standard deviation (s_D=1.1):

$$t = \frac{2.5}{1.1/\sqrt{10}} \approx 7.18$$

```
from scipy.stats import t

# Given values
t_value = 7.18
df = 58  # degrees of freedom (n_1 + n_2 - 2)

# Calculate the two-tailed p-value
p_value = 2 * (1 - t.cdf(t_value, df))

print(f"P-value: {p_value:.4f}") #0.000
```

p-value is less than 0.05, there exists difference in weight before and after diet.

11.2 ANOVA (Analysis of Variance) Explanation

ANOVA (Analysis of Variance) is a statistical method used to compare the means of **three or more groups** to determine if there are statistically significant differences between them. It does this by analyzing the variance within groups and between groups.

- **Null Hypothesis (H_0)**: All group means are equal (no significant difference).
- **Alternative Hypothesis (H_1)**: At least one group mean is significantly different.
- **Total Sum of Squares (SST)**:

  $SST = \sum_{i=1}^{k} \sum_{j=1}^{n} (X_{ij} - \bar{X})^2$
- **Sum of Squares Between Groups (SSB)**:

  $SSB = \sum_{i=1}^{k} n_i (\bar{X}_i - \bar{X})^2$
- **Sum of Squares Within Groups (SSW)**:

  $SSW = \sum_{i=1}^{k} \sum_{j=1}^{n} (X_{ij} - \bar{X}_i)^2$

Suppose we want to analyze the scores of students from three different schools to see if there is a significant difference in scores between these schools. We have the following data:

School	Student Scores
A	80, 85, 90
B	70, 75, 80
C	95, 100, 105

Let's calculate the mean score for each school and the overall mean score.

- **School A Mean**:

$$\bar{X}_A = \frac{80 + 85 + 90}{3} = 85$$

- **School B Mean**:

$$\bar{X}_B = \frac{70 + 75 + 80}{3} = 75$$

- **School C Mean**:
$$\bar{X}_C = \frac{95 + 100 + 105}{3} = 100$$
- **Overall Mean**:
$$\bar{X} = \frac{(80 + 85 + 90) + (70 + 75 + 80) + (95 + 100 + 105)}{9} = \frac{850}{9} \approx 86.67$$

11.2.1 Total Sum of Squares (SST)

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n} (X_{ij} - \bar{X})^2$$

Let's calculate SST for our example:

- For School A:
  - $$(80 - 86.67)^2 + (85 - 86.67)^2 + (90 - 86.67)^2 = 58.33$$
- For School B:
  - $$(70 - 86.67)^2 + (75 - 86.67)^2 + (80 - 86.67)^2 = 458.33$$
- For School C:
  - $$(95 - 86.67)^2 + (100 - 86.67)^2 + (105 - 86.67)^2 = 583.33$$
- $$SST = 58.33 + 458.33 + 583.33 = 1100$$

11.2.2 Sum of Squares Between Groups (SSB)

$$SSB = \sum_{i=1}^{k} n_i (\bar{X}_i - \bar{X})^2$$

- For School A:
  - $$3 \times (85 - 86.67)^2 = 3 \times (-9.44)^2 = 3 \times 89.03 = 8.33$$

- For School B:
  - $$3 \times (75 - 86.67)^2 = 3 \times (-19.44)^2 = 3 \times 378.03 = 408.33$$
- For School C:
  - $$3 \times (100 - 86.67)^2 = 3 \times (5.56)^2 = 3 \times 30.81 = 533.33$$
- $$SSB = 8.33 + 408.33 + 533.33 = 950$$

11.2.3 Sum of Squares Within Groups (SSW)

$$SSW = \sum_{i=1}^{k} \sum_{j=1}^{n} (X_{ij} - \bar{X}_i)^2$$

- For School A:
  - $$(80-85)^2+(85-85)^2+(90-85)^2 = (-5)^2+0^2+5^2 = 25+0+25 = 50$$
- For School B:
  - $$(70-75)^2+(75-75)^2+(80-75)^2 = (-5)^2+0^2+5^2 = 25+0+25 = 50$$
- For School C:
  - $$(95-100)^2+(100-100)^2+(105-100)^2 = (-5)^2+0^2+5^2 = 25+0+25 = 50$$
- $$SSW = 50 + 50 + 50 = 150$$

The relationship between these sums of squares is given by:

$$SST = SSB + SSW$$

In our example:$ SST = 1100 $, $ SSB = 950 $ and $ SSW = 150 $. Thus, $ 1100 = 950 + 150 $ confirming the relationship.

11.3 Chi-Square Test of Independence

The Chi-square test helps determine if there is a significant association between two categorical variables.

- **Null Hypothesis (H_0)**: The two categorical variables are independent.
- **Alternative Hypothesis (H_1)**: The two categorical variables are NOT independent.

The contingency table can be represented as:

$$
\begin{array}{c|c|c|c|c}
 & Y_1 & Y_2 & \cdots & Y_m \\
\hline
X_1 & n_{11} & n_{12} & \cdots & n_{1m} \\
X_2 & n_{21} & n_{22} & \cdots & n_{2m} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
X_k & n_{k1} & n_{k2} & \cdots & n_{km}
\end{array}
$$

1. Compute the expected frequency E_{ij} for each cell:

$$E_{ij} = \frac{n_{i\cdot} \times n_{\cdot j}}{N}$$

Where N is the total number of observations.

2. Calculate the Chi-square statistic:

$$\chi^2 = \sum_{i=1}^{k} \sum_{j=1}^{m} \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$$

3. Compare the calculated χ^2 value with the critical value from the Chi-square distribution with ((k-1)(m-1)) degrees of freedom at a chosen significance level (α).

Numerical Example

Let's say we survey 100 individuals to analyze the relationship between gender (Male, Female) and preference for a product type (A, B). Here is the cross-tabulation:

	A	B	$Total$
Male	30	20	50
Female	20	30	50
Total	50	50	100

Using the above table:

1. $E_{11} = \frac{50 \times 50}{100} = 25$
2. $E_{12} = \frac{50 \times 50}{100} = 25$
3. Calculate χ^2:

$$\chi^2 = \frac{(30-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(30-25)^2}{25} = 4$$

Given $df = (2-1)(2-1) = 1$ and a significance level of 0.05, the critical value from the Chi-square table is 3.841. Since $4 > 3.841$, we reject the null hypothesis and conclude that there is a significant relationship between gender and product preference.